

Botochtone: Culturally Adapted Transformer-Based Moderation for Indigenous Communities

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Abstract. Online platforms have become essential spaces for Indigenous communities, yet they also expose members to culturally targeted harassment, historical denialism, and discriminatory content that generic moderation systems often fail to recognize. While Transformer-based moderation has achieved strong performance on general harmful-content detection tasks, culturally adapted moderation for Indigenous online communities remains relatively underexplored. This paper presents **Botochtone**, a Transformer-based moderation framework for Indigenous online safety. It adapts a Ministral-8B instruction-tuned Transformer through Low-Rank Adaptation (LoRA), Indigenous-aware prompt engineering, and a purpose-built Indigenous-oriented dataset. The dataset is grounded in publicly available Canadian reports and resources on Indigenous online safety, and the study focuses specifically on the Canadian Indigenous context. Through an incremental ablation study, the final system reaches an F1-score of 98.3% on an Indigenous-oriented evaluation benchmark. Among the evaluated configurations, the largest performance improvement is observed when Indigenous-oriented training examples are incorporated. We also evaluate the final system against a grey-box adversarial LLM that rephrases harmful messages while attempting to evade detection. After human validation of the generated rewrites, the semantically validated attack success rate is 8.6%.

Keywords: Content moderation · Indigenous online safety · Transformer-based moderation · Cyberbullying detection · Cultural adaptation

1 Introduction

Online social platforms have become essential spaces for communication, community building, and cultural expression. However, they also expose users to cyberbullying, hate speech, manipulation, and other forms of harmful content. Indigenous communities are particularly vulnerable to online harm due to the

persistence of historical discrimination, cultural marginalization, and targeted attacks on Indigenous identities and traditions.

Research has shown that Indigenous youth experience disproportionately high levels of cybervictimization compared with non-Indigenous populations [10]. Beyond conventional cyberbullying, Indigenous individuals frequently encounter culturally specific forms of harmful content, including attacks on Indigenous identity, denial of historical injustices, mockery of traditional practices, and challenges to Indigenous sovereignty and cultural heritage [23]. Such forms of harm are often overlooked by existing moderation systems, which are primarily trained on generic datasets and broad definitions of harmful behavior.

Recent advances in Large Language Models (LLMs) and Transformer architectures [29,27,14] have significantly improved automated content moderation through stronger semantic and contextual understanding. Nevertheless, the effectiveness of these models remains dependent on the quality and representativeness of the data used for adaptation. While Transformer-based moderation has been widely studied for general harmful-content detection, culturally adapted moderation systems and Indigenous-oriented moderation datasets remain relatively underexplored in the literature.

This paper presents **Botochtone**⁴, a Transformer-based moderation framework designed to support Indigenous online safety. It combines a locally deployed Ministral-8B instruction-tuned Transformer with Low-Rank Adaptation (LoRA) [12], Indigenous-aware prompt engineering, and a purpose-built Indigenous-oriented dataset derived from Canadian Heritage Indigenous online safety reports [3].

The main contributions of this work are as follows. Building on prior work showing the value of community- and context-specific information for content moderation, this study focuses on an Indigenous context that remains under-represented in current moderation research. Specifically, Botochtone provides a culturally adapted moderation framework, an Indigenous-oriented dataset grounded in publicly available Canadian resources, and an empirical evaluation of this approach in the Canadian Indigenous context.

- We design and implement Botochtone, a Transformer-based moderation framework integrating LoRA adaptation, culturally informed prompting, and Indigenous-oriented training data for cyberbullying and harmful-content detection.
- We construct a purpose-built Indigenous-oriented moderation dataset by combining publicly available harmful-content corpora with culturally adapted examples derived from publicly available Canadian reports and resources on Indigenous online safety.
- We conduct an ablation study to quantify the contribution of LoRA adaptation, Indigenous-aware prompting, and culturally representative training data.

⁴ The name *Botochtone* is a blend of “bot” and the French *autochtone* (“Indigenous”).

- We show that the largest performance improvement among the evaluated configurations occurs when Indigenous-oriented training examples are incorporated, with the final configuration achieving an F1-score of 98.3% on the evaluation benchmark.
- We assess the final detector against a grey-box adversarial language model that attempts to rephrase harmful messages while preserving their meaning and evading detection. We report both detector-level bypasses and semantically validated evasions based on independent human annotation.
- We show that effective Transformer-based moderation can be deployed entirely on local hardware using LoRA and 4-bit quantization, supporting privacy-preserving moderation for Indigenous organizations and communities.

The remainder of this paper is organized as follows. Section 2 presents the theoretical background. Section 3 reviews related work. Section 4 describes the proposed approach, dataset construction process, and adaptation methodology. Section 5 presents the empirical evaluation. Section 6 discusses the findings and their implications. Section 7 addresses threats to validity. Finally, Section 8 concludes the paper.

2 Background

2.1 Indigenous Online Harm

Indigenous communities are disproportionately exposed to cybervictimization, cyberbullying, and culturally specific forms of online harm, including attacks on Indigenous identity, denial of historical injustices, discriminatory stereotypes, and harassment targeting traditional practices [10,23,3]. Unlike conventional cyberbullying, these forms of harm often rely on contextual references, historical narratives, and culturally encoded language that are difficult to detect using generic moderation systems. As a result, moderation approaches trained exclusively on general-purpose datasets may fail to recognize community-specific harmful content [2].

2.2 Transformer Adaptation and LoRA

Transformer architectures [29] constitute the foundation of modern Large Language Models (LLMs), providing strong contextual reasoning capabilities for tasks such as cyberbullying detection and content moderation. Instruction-tuned models such as Ministral-8B [19] can be adapted to specialized domains through prompt engineering and parameter-efficient fine-tuning techniques.

Low-Rank Adaptation (LoRA) [12] is a widely adopted parameter-efficient fine-tuning method that introduces trainable low-rank matrices alongside frozen pretrained weights:

$$W' = W_0 + \Delta W = W_0 + BA, \quad r \ll \min(d, k), \quad (1)$$

where W_0 denotes the pretrained weights and BA represents the trainable low-rank update. This approach enables domain adaptation while updating only a small fraction of model parameters. In Botochtone, LoRA is combined with Indigenous-aware prompting to incorporate both learned and contextual knowledge during moderation.

3 Related Work

Cyberbullying and Harmful Content Detection. Automated harmful-content detection has evolved from traditional machine learning approaches based on handcrafted linguistic features and shallow classifiers [15] to Transformer-based architectures capable of capturing contextual and semantic relationships within text [29,25,24]. Recent advances have substantially improved the detection of cyberbullying, hate speech, toxicity, and online harassment. Industrial systems such as Perspective API [16] contributed to the large-scale deployment of automated toxicity detection, although they were primarily developed using broad, population-level definitions of harmful content. Despite these advances, most existing moderation models remain optimized for generic datasets and may struggle to recognize culturally contextualized forms of harm affecting specific communities.

Datasets for Harmful Content Moderation. The performance of modern moderation systems is closely tied to the quality and representativeness of their training data. HateXplain [17] introduced an influential benchmark combining hate-speech annotations with human rationales, supporting both classification and explainability. Vidgen *et al.* [30] demonstrated that dynamically generated and adversarially constructed datasets can significantly improve robustness when standard benchmarks fail to cover the target domain. A complementary line of work shows that LLM-based synthetic data augmentation can improve detection in low-resource hate-speech scenarios, although it raises misuse and governance concerns that call for careful oversight [8]. While numerous datasets exist for cyberbullying, hate speech, and toxicity detection, most focus on generic manifestations of harmful content and rarely capture culturally specific forms of discrimination. As a result, datasets designed for Indigenous online safety remain largely absent from the current moderation landscape.

LLM-Based Safety Classifiers. Recent work has increasingly leveraged instruction-tuned Large Language Models (LLMs) for content moderation. Systems such as Llama Guard [13], WildGuard [9], and ShieldGemma [32] formulate safety assessment as an instruction-following task and achieve strong performance across multiple moderation benchmarks. Beyond sentence-level toxicity, Mendu *et al.* [18] propose a long-form harmful-content filter and report that a substantial fraction of outputs from state-of-the-art LLMs remain unsafe, underscoring the need for dedicated safety layers. However, these systems are designed around

broad safety taxonomies and are not specifically adapted to the cultural contexts of marginalized communities.

Culturally Adapted and Community-Specific Moderation. A growing body of work shows that community-specific adaptation substantially improves moderation effectiveness. Mozafari *et al.* [20] revealed systematic racial bias in BERT-based hate classifiers and proposed reweighting strategies to mitigate it, illustrating how generic models can disproportionately misclassify dialectal or minority language. Consistent with this, Hartmann *et al.* [11] audited five commercial moderation APIs and found that they simultaneously over-moderate identity terms and counterspeech while under-moderating implicit hate—a strong argument against deploying off-the-shelf moderation for vulnerable communities without local auditing. Blodgett *et al.* [4] similarly argue that NLP systems trained on generic corpora often fail to adequately represent marginalized populations.

To address these gaps, recent systems explicitly incorporate cultural knowledge. Muhammad *et al.* [21] introduced AfriHate, a multilingual collection of hate-speech datasets for African languages, demonstrating that moderation quality improves markedly when data are constructed with native speakers familiar with local culture. Ye *et al.* [31] proposed a federated few-shot approach that combines culture-specific data with privacy-preserving local training for marginalized communities. Park *et al.* [22] showed that retrieving cultural context substantially raises cross-cultural moderation accuracy, while Gajewska *et al.* [6] formalized identity-aware moderation through culturally grounded community agents. Earlier work by Arslan *et al.* [2] likewise demonstrated that Transformer-based models benefit from contextual knowledge when detecting cyberbullying targeting LGBTQ+ individuals.

Despite this progress, to the best of our knowledge no prior work has investigated Transformer-based moderation specifically adapted to Indigenous online communities. Existing research on Indigenous online safety has largely emerged from policy, social science, and public-health perspectives [3,10,23], with comparatively limited attention devoted to AI-driven moderation solutions. Botochtone addresses this gap by combining a culturally curated Indigenous-oriented dataset, Indigenous-aware prompting, and privacy-preserving local deployment within a single framework. To the best of our knowledge, there is currently no shared Indigenous-oriented moderation benchmark that would support a controlled comparison with these approaches. Existing community-specific systems have been evaluated on different populations, datasets, languages, and task formulations, making their reported results difficult to compare directly with Botochtone. We therefore evaluate the contribution of the different adaptation components under the same experimental conditions using a common Indigenous-oriented benchmark. The dataset and benchmark introduced in this work also provide an initial resource that can support future comparisons of moderation approaches in the Canadian Indigenous context.

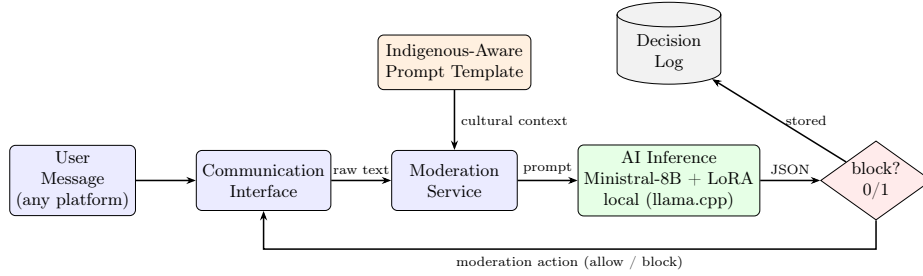


Fig. 1: End-to-end moderation process in Botochtone. A user message from any communication platform is captured by the communication interface, enriched with the Indigenous-aware prompt template by the moderation service, classified by the locally deployed Ministral-8B model, and returned as a binary decision that is both enforced and logged for auditing.

Parameter-Efficient Transformer Adaptation. Parameter-efficient fine-tuning methods have become a standard strategy for adapting large Transformer models to specialized domains. Among these approaches, Low-Rank Adaptation (LoRA) [12] has emerged as one of the most widely adopted techniques due to its ability to achieve competitive performance while updating only a small fraction of model parameters. LoRA has been successfully applied to foundation models such as LLaMA [27] and Mistral/Ministral [14,19] across a wide range of NLP tasks. These advances suggest that combining parameter-efficient adaptation with culturally representative training data and community-specific prompting offers a promising direction for developing moderation systems tailored to underrepresented populations.

4 Proposed Approach

This section presents the Botochtone moderation framework. It describes the system architecture, the construction of the Indigenous-oriented dataset, the LoRA-based adaptation of the Transformer model, the Indigenous-aware prompt engineering, and the privacy-preserving local deployment pipeline.

4.1 System Architecture

Botochtone follows a modular, four-layer architecture, illustrated in Figure 1 and described below in the order a message traverses the framework⁵.

Communication layer. A user message is first captured by the communication interface. In our prototype, this layer is implemented as a Discord bot using `discord.py`, but it acts only as a platform adapter: the message is immediately forwarded as raw text to the moderation service.

⁵ The prototype code and the Indigenous-oriented dataset are available at <https://github.com/OulmalmeChaimaa/Botochtone>.

Table 1: Public harmful-content datasets: raw sizes and subsampled training contributions.

Dataset	Raw size	Training contrib. Purpose
HateXplain [17]	20,109	476 Hate speech
FTR [28]	2,856	99 Racism detection

Moderation layer. The moderation service, exposed through a FastAPI endpoint, wraps the incoming message with the Indigenous-aware prompt template (described in Section 4.5) to produce the structured prompt submitted to the model.

AI inference layer. The prompt is processed by the locally deployed, LoRA-adapted Ministral-8B Transformer running through `llama.cpp`. The model returns a binary moderation decision encoded as JSON.

Data management layer. The decision is returned to the communication interface, which enforces the corresponding action (allow or block), and is simultaneously logged in a PostgreSQL database for auditing and later analysis.

Because the communication layer is a thin platform adapter, the architecture can be integrated into other text-based communication environments by replacing this single layer. In the current prototype, Discord serves as the integration surface, while the moderation model itself operates directly on the text received by the moderation service.

4.2 Base Transformer Model: Ministral-8B

The moderation engine is built on `mistralai/Ministral-8B-Instruct-2410` [19], a decoder-only Transformer model comprising approximately 8 billion parameters. The model employs Grouped-Query Attention (GQA) [1] with 32 attention heads and 8 key-value heads, a sliding-window attention mechanism, 32 Transformer blocks, a hidden dimension of 4096, and a feed-forward network dimension of 14336. These architectural characteristics enable strong contextual reasoning while remaining compatible with local deployment and parameter-efficient adaptation strategies.

4.3 Dataset Construction

Dataset construction is a primary scientific contribution of this work. To the best of our knowledge, publicly available moderation datasets rarely capture Indigenous-directed forms of online harm, motivating the construction of a culturally representative corpus.

Public Harmful-Content Datasets Two publicly available datasets were incorporated.

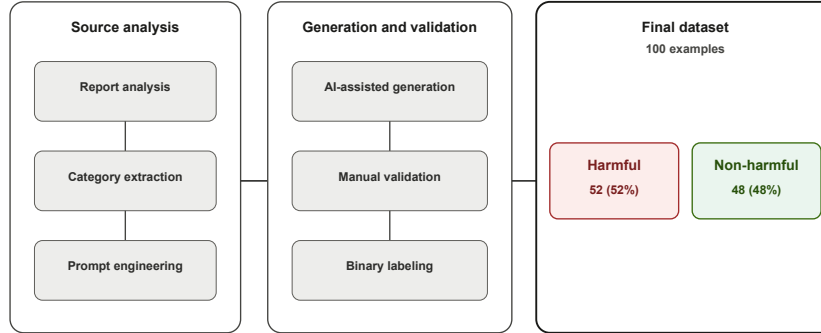


Fig. 2: Indigenous-oriented dataset generation pipeline.

HateXplain [17] provides hate-speech annotations with human rationales and was used in this study as a source of labeled harmful-content examples. **FTR** (a dataset of French tweets annotated for racist language) [28] provides French-language examples of racist and discriminatory content that complement the broader harmful-content categories represented in HateXplain. Both datasets were subsampled via systematic interval sampling before inclusion in the training pipeline, yielding 476 HateXplain and 99 FTR examples.

Indigenous-Oriented Dataset: Generation Pipeline The core scientific contribution of the dataset construction is the Indigenous-oriented harmful-content corpus. Figure 2 presents the seven-stage generation pipeline. To the best of our knowledge, this is the first purpose-built Indigenous-oriented dataset for content moderation. The dataset was developed from publicly available Canadian reports and resources documenting Indigenous online harms, with the *What We Heard Report: Indigenous Online Safety* [3] serving as the main source for identifying the forms of harm represented in the corpus. Accordingly, this first dataset is intentionally limited to the Canadian Indigenous context. Its size reflects the scope and level of detail available in the public sources used in this study. Extending the dataset to Indigenous communities in other countries would require identifying appropriate community and country specific sources and considering their cultural context, access conditions, and data governance requirements.

Step 1 – Report analysis. The *What We Heard Report: Indigenous Online Safety* published by Canadian Heritage [3] was analyzed to identify recurring forms of online harm directed toward Indigenous peoples.

Step 2 – Category extraction. Six categories of culturally specific harmful content were systematically extracted (Table 2).

Table 2: Indigenous-oriented harmful content categories derived from [3].

Category	Description
Identity-based discrimination	Attacks targeting Indigenous identity and belonging
Cultural discrimination	Mockery or denial of traditional practices and values
Historical denialism	Denial of Residential Schools, treaties, injustices
Indigenous rights attacks	Delegitimization of sovereignty and treaty rights
Negative stereotypes	Reinforcement of discriminatory stereotypes
Economic prejudice	Misinformation about Indigenous economic support

Step 3 – Prompt engineering. Structured prompts were designed for each category, grounding generation in specific harmful narratives documented in the report.

Step 4 – AI-assisted generation. A conversational AI system was prompted to generate representative examples for each category. Approximately 150 candidate examples were generated at this stage.

Step 5 – Manual validation. All generated examples were manually reviewed by members of the research team. Examples that were ambiguous, factually inaccurate, or insufficiently representative of the targeted forms of online harm were discarded, yielding 100 retained examples.

Step 6 – Binary labeling. Retained examples were labeled according to a binary moderation scheme, where harmful content was assigned the label 1 and non-harmful content the label 0.

Step 7 – Final dataset. The resulting Indigenous-oriented dataset contains **100 examples**: 52 harmful (52%) and 48 non-harmful (48%), distributed across the six categories.

Dataset Quality Assessment. Beyond manual review, we quantitatively assessed the generated dataset to verify that AI-assisted generation did not yield repetitive or low-diversity content, a common concern with synthetic data. The corpus exhibits high lexical diversity, with a type-token ratio of 0.50 over 1,048 tokens (526 unique). Semantic redundancy is low: the average pairwise TF-IDF cosine similarity between examples is 0.04 and the maximum is 0.46. No pair exceeded the selected 0.6 similarity threshold, providing quantitative evidence of lexical diversity and low redundancy alongside the manual review performed by the research team.

Dataset Characteristics. The Indigenous-oriented dataset consists of 100 short, written messages in French, with an average length of 10.5 words (range: 6 to 14 words). Each example is an independent social media style message rather than part of a multi-turn conversation, and the dataset does not contain transcriptions of spoken language. The final set contains 52 harmful and 48 non-harmful examples.

Table 3: Final training corpus: exact composition and class distribution.

Source	Examples	Harmful	Non-harmful
HateXplain (subsampling)[17]	476	299 (62.8%)	177 (37.2%)
FTR (subsampling)[28]	99	42 (42.4%)	57 (57.6%)
Indigenous-oriented dataset	100	52 (52.0%)	48 (48.0%)
Total	675	393 (58.2%)	282 (41.8%)

The complete training corpus is bilingual. The two public datasets complement the Indigenous-oriented examples: FTR [28] contains French tweets, whereas HateXplain [17] contains English social media posts. The resulting training corpus therefore combines 476 English HateXplain examples with 199 French examples from FTR and the Indigenous-oriented dataset. The 132-message evaluation benchmark also contains both French and English messages and follows the same single-message format.

The Indigenous-oriented examples were manually reviewed by the research team before inclusion in the final dataset.

Final Training Corpus. Table 3 presents the exact composition of the final training corpus.

Evaluation Benchmark To measure moderation performance, we constructed a separate Indigenous-oriented evaluation benchmark following the same procedure as the training dataset (Figure 2): messages were generated with AI assistance from the six harm categories derived from the Canadian Heritage report [3] and then manually reviewed, but using *distinct message instances* held out from training. The benchmark consists of 132 Indigenous-oriented messages: **60 harmful** and **72 non-harmful** (45.5% / 54.5%). No benchmark example overlaps with the training corpus (HateXplain, FTR, or the Indigenous-oriented training set).

The benchmark was used consistently to evaluate all model configurations under identical experimental conditions, providing a common basis for comparing successive model versions and measuring the impact of architectural, prompting, and dataset-related modifications. The slight class imbalance motivates the use of F1-score as the primary metric, complemented by Precision, Recall, and Accuracy.

4.4 LoRA-Based Transformer Adaptation

LoRA [12] was applied to specialize the Ministral-8B Transformer for harmful-content moderation. LoRA was selected because it enables efficient adaptation of large Transformer models while maintaining a computational footprint compatible with local deployment.

Table 4: LoRA adaptation configuration.

Parameter	Value
Rank (r)	16
Alpha (α)	32
Dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Trainable parameters	$\approx 42\text{M}$ ($<0.6\%$ of model)
Learning rate	$2\text{e-}4$
Epochs	3
Batch size	2
Gradient accumulation steps	8 (effective batch size: 16)
Maximum sequence length	256
Numerical precision	bf16

Low-rank decomposition matrices were inserted alongside all attention projection layers ($\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$, $\mathbf{O}$) and feed-forward network layers (gate, up, down) in each Transformer block. With rank $r = 16$, LoRA introduces approximately **42 million trainable parameters**, representing less than **0.6%** of the total model parameters. Table 4 summarizes the complete adaptation configuration.

Training examples were formatted according to the Ministral instruction template, consisting of a system instruction, a user message, and the expected moderation decision encoded as a JSON response. During training, only the moderation decision contributed to the optimization objective, ensuring that the model learned the harmful-content classification task rather than reproducing the prompt itself. The model was trained for three epochs using fixed hyperparameters and without hyperparameter search.

4.5 Indigenous-Aware Prompt Engineering

At inference time, a culturally informed system prompt provides the Transformer with structured contextual knowledge:

- **Historical sensitivity:** Recognition of high-sensitivity topics including the Residential School system, the Indian Act, the Sixties Scoop, and Missing and Murdered Indigenous Women and Girls (MMIWG).
- **Coded language detection:** Vigilance for indirect expressions and contextual narratives commonly used to delegitimize Indigenous rights and communities.
- **Sovereignty and identity:** Consideration of Indigenous self-identification, treaty rights, and sovereignty-related discussions.
- **Intent versus impact:** Recognition that harmful content may occur even in the absence of explicit offensive language.
- **Cultural reclamation exception:** Protection of legitimate cultural expression, storytelling, advocacy, and identity-sharing activities.

This prompt is applied at each inference call, enabling the Transformer to interpret messages against a cultural and historical framework without requiring retraining.

4.6 Deployment and Inference

Following LoRA adaptation, the model weights were merged with the base Ministral-8B weights and exported to the GGUF format. Four-bit mixed quantization (Q4_K_M) was then applied through `llama.cpp` [7], reducing memory requirements from approximately 16 GB to approximately 5 GB while maintaining strong moderation performance.

Inference operates with a temperature of 0.01 to maximize decision consistency and minimize output variability. Model inference is performed entirely on local infrastructure, and message content is not sent to an external inference API after being received by the moderation service.

This design can support privacy and Indigenous data-sovereignty objectives by reducing dependence on external AI inference services and keeping moderation inference under local control.

5 Empirical Evaluation

This section presents the empirical evaluation of Botochtone. It describes the experimental setup and evaluation metrics, then reports an incremental ablation study that quantifies the contribution of LoRA adaptation, Indigenous-aware prompting, and culturally representative training data to overall moderation performance.

5.1 Experimental Setup

All configurations were evaluated on the same Indigenous-oriented evaluation benchmark composed of 132 messages (60 harmful and 72 non-harmful). No benchmark examples overlap with the training corpus. To support reproducibility, the data, prompts, experimental scripts, and results are publicly available. The repository also includes a README with instructions for using the released resources and reproducing the experiments: <https://github.com/OulmalmeChaimaa/Botochtone>.

5.2 Evaluation Metrics

Four complementary metrics, standard for classification tasks [26], were computed from the confusion matrix of each configuration:

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP}, & \text{Recall} &= \frac{TP}{TP + FN}, \\ \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN}, & \text{F1} &= 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \end{aligned}$$

Table 5: Performance of the four Botochtone configurations on the fixed Indigenous-oriented evaluation benchmark.

Config.	Description	Precision (%)	Recall (%)	Accuracy (%)	F1 (%)
v0.2	Baseline (no adaptation)	72.3	100.0	82.6	83.9
v3	+ LoRA on generic corpus	100.0	76.7	89.4	86.8
v5	+ Indigenous-aware prompt	100.0	73.3	87.9	84.6
v8	+ Indigenous-oriented data (final)	100.0	96.7	98.5	98.3

Precision measures the proportion of flagged messages that are truly harmful, Recall measures the proportion of harmful messages correctly detected, Accuracy measures overall classification performance, and F1-score provides a balanced assessment of Precision and Recall.

5.3 Ablation Study: Model Configurations

Four configurations were evaluated in an incremental ablation study, each introducing one additional adaptation component:

- **v0.2 — Baseline.** Unadapted Ministral-8B with a generic moderation prompt and no LoRA training.
- **v3 — LoRA + Generic Data.** LoRA adaptation on a generic harmful-content corpus (HateXplain + FTR, subsampled), with the generic prompt.
- **v5 — LoRA + Indigenous Prompt.** Same LoRA-adapted model as v3, with the Indigenous-aware cultural prompt applied at inference time.
- **v8 — Full Configuration.** LoRA adaptation on the complete training corpus including Indigenous-oriented examples, with the Indigenous-aware prompt (final Botochtone system).

Table 5 reports the full evaluation results. Figure 3 compares the F1-score across configurations. As shown in Figure 3, the largest performance improvement occurs when Indigenous-oriented training examples are incorporated into the adaptation process.

Baseline: Unadapted Ministral-8B (v0.2). Figure 4a presents the confusion matrix for the unadapted Ministral-8B baseline. Without domain-specific training, the model correctly identifies all 60 harmful messages (Recall = 100%) but simultaneously flags 23 of 72 non-harmful messages as harmful (FP = 23, Precision = 72.3%). The resulting F1-score of 83.9% reflects a moderation system that is functional but too aggressive: a 32% false-positive rate would likely limit its suitability for deployment in real-world community moderation settings.

Effect of LoRA Adaptation on Generic Corpora (v3). LoRA adaptation on the generic corpus (HateXplain + FTR) eliminates false positives entirely, raising Precision from 72.3% to 100%. However, the adaptation reduces Recall

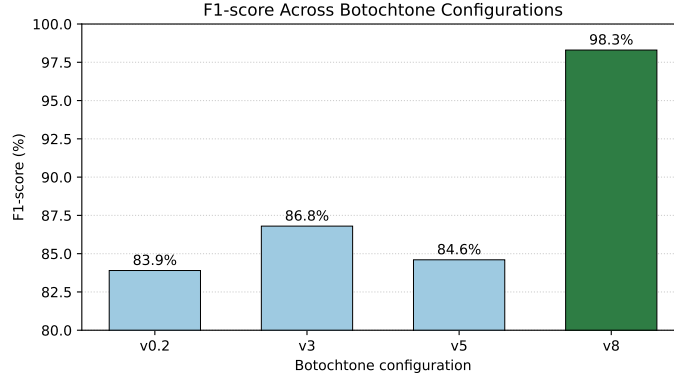


Fig. 3: F1-score across Botochtone configurations.

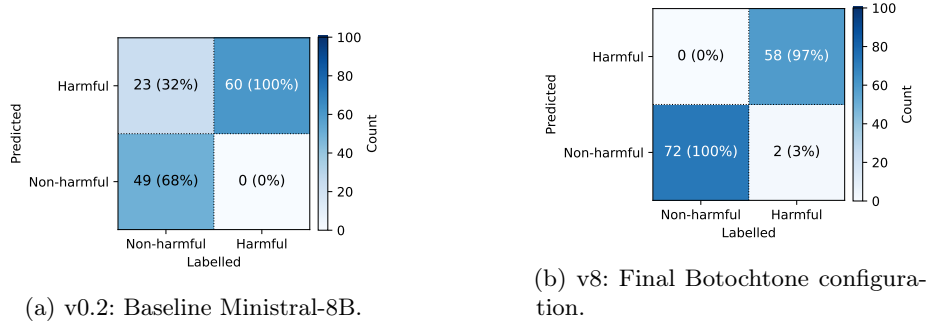


Fig. 4: Confusion matrices comparing the baseline (v0.2) and final (v8) configurations on the Indigenous-oriented evaluation benchmark.

from 100% to 76.7% (FN = 14), and the overall F1 improves only marginally to 86.8%. This result shows that adaptation on the generic corpora alone does not provide the same level of recall obtained by the final Indigenous-oriented configuration. Because the benchmark does not contain per-category annotations, we do not attribute the missed cases to specific forms of Indigenous-directed harm.

Effect of Indigenous-Aware Prompt Engineering (v5). Introducing the Indigenous-aware cultural prompt reduces Recall marginally from 76.7% to 73.3% (F1: 86.8% → 84.6%), while Precision remains at 100% (zero false positives). This result suggests that prompt-based cultural adaptation alone may be insufficient to improve detection performance without culturally representative training examples.

Effect of Indigenous-Oriented Training Data (v8 — Final System).

The most significant performance improvement is achieved by incorporating Indigenous-oriented training examples into the LoRA adaptation. With the complete training corpus and Indigenous-aware prompt, Recall increases from 73.3% to 96.7% (FN reduced from 16 to 2), while Precision remains at 100%. The final F1-score reaches 98.3% and overall Accuracy reaches 98.5%.

Figure 4b presents the confusion matrix for the final configuration. Only 2 of 60 harmful messages are missed, and no non-harmful message is incorrectly blocked.

The contrast with v3 is particularly instructive: v8 achieves substantially higher Recall (96.7% vs. 76.7%). Among the evaluated configurations, the largest observed improvement occurs between v5 and v8, when Indigenous-oriented examples are incorporated into LoRA adaptation.

Figure 4 visually illustrates the reduction in both false positives and false negatives achieved by the final configuration compared with the baseline model.

5.4 Adversarial Robustness Evaluation

We evaluated whether an adversarial language model could rephrase harmful messages so that they bypass the final Botochtone detector while preserving their original harmful meaning. We use a grey-box setting in which the attacker receives a natural-language description of the moderation policy but has no access to the detector weights, its exact system prompt, or detector feedback. The attacker is an instruction-tuned Ministral-8B model. For each original message, it generates three candidate rewrites: one using greedy decoding and two using sampling.

The attack set contains the 58 harmful messages correctly detected by the final system at baseline. The two baseline false negatives are excluded because they already bypass the detector without adversarial modification. The attacker generated 174 candidate rewrites and produced no refusals. Among these candidates, 19 were not blocked by Botochtone, corresponding to 12 distinct original messages. The resulting raw message-level evasion rate is therefore $12/58 = 20.7\%$.

Detector bypass alone does not establish a successful attack because a rewrite may evade detection by changing or removing the original harmful meaning. We therefore manually evaluated the 19 non-blocked candidates, which originated from 12 distinct messages. Two annotators independently assessed whether each rewrite preserved the original target, harmful meaning, and hostile intent. They agreed on 17 of the 19 cases (89.5%), corresponding to a Cohen’s kappa [5] of 0.79 before adjudication. The two disagreements were resolved through discussion between the annotators until consensus was reached. After semantic validation and adjudication, only five of the 12 original messages with a raw detector bypass had at least one non-blocked rewrite that preserved the original harmful meaning and intent. For the other seven messages, none of the non-blocked rewrites satisfied the semantic-preservation criterion. Therefore, only five of the 58 attacked messages resulted in a semantically valid evasion, corresponding to a validated attack success rate (ASR) of 8.6%.

Table 6: Adversarial evaluation of the final system under the grey-box, best-of-three protocol.

Metric	Value
Baseline-detected harmful messages	58 / 60
Candidate rewrites (refusals)	174 (0)
Originals with ≥ 1 non-blocked rewrite	12 / 58 = 20.7%
Inter-annotator agreement	17 / 19 = 89.5% ($\kappa = 0.79$)
Originals with valid evasions	5 / 58
Validated ASR	8.6%

Table 6 summarizes the main results of the adversarial evaluation. The difference between the raw evasion rate (20.7%) and the validated ASR (8.6%) shows the importance of checking semantic preservation when evaluating adversarial rephrasing. Under the tested protocol, 53 of the 58 baseline-detected harmful messages did not produce a semantically valid evasion. These results are specific to the grey-box, single-attacker, best-of-three setting considered here and do not establish robustness against other adversarial strategies.

6 Discussion

6.1 Cultural Adaptation and Indigenous-Oriented Data

The empirical evaluation shows that the largest performance improvement among the evaluated configurations occurs when Indigenous-oriented training examples are incorporated. While LoRA adaptation and Indigenous-aware prompting modify the moderation behaviour of the model, the transition from v5 to v8 increases Recall from 73.3% to 96.7% while maintaining 100% Precision. These results suggest that Indigenous-oriented training examples provide useful information that is not sufficiently represented in the generic corpora used in this study.

This result should be interpreted within the scope of the current benchmark. Because the evaluation data provide binary harmful/non-harmful labels but no annotations for individual harm categories, the experiment does not establish which specific forms of Indigenous-directed harm account for the observed improvement.

6.2 Indigenous-Specific Harm and Detection Behaviour

The six harm categories represented in the Indigenous-oriented corpus (identity-based discrimination, cultural discrimination, historical denialism, attacks on Indigenous rights, negative stereotypes, and economic prejudice) include forms of harm that can depend on historical or cultural context rather than explicit slurs. This distinction is relevant to the experimental results. The configuration trained on generic corpora (v3) achieves 76.7% recall, and adding the Indigenous-aware

prompt (v5) results in 73.3% recall. In contrast, the final configuration, which also incorporates Indigenous-oriented training examples (v8), reaches 96.7% recall. Among the evaluated configurations, the largest improvement therefore occurs when Indigenous-oriented examples are included in training, suggesting that culturally representative data can help the model recognize forms of harm that are underrepresented in generic moderation corpora.

The current evaluation benchmark provides binary harmful/non-harmful labels but does not include annotations for the six individual harm categories. We therefore report aggregate performance rather than per-category results. Developing a category-annotated benchmark would allow future work to examine which forms of Indigenous-directed harm benefit most from culturally adapted training.

6.3 Robustness to Adversarial Rephrasing

The adversarial evaluation shows that Botochtone can be bypassed by meaning-preserving rephrasing, although successful evasions were limited under the protocol considered here. At the message level, 12 of the 58 baseline-detected harmful messages produced at least one non-blocked rewrite, corresponding to a raw evasion rate of 20.7%. After human assessment of semantic preservation, only five original messages retained at least one valid evasion, reducing the validated attack success rate to 8.6%.

This difference highlights the importance of semantic validation in adversarial moderation experiments: a rewrite that is no longer detected is not necessarily a successful attack if it no longer preserves the target, harmful meaning, and intent of the original message. The 8.6% result should therefore be interpreted within the scope of this experiment, which considers a single attacker model, a grey-box setting, a best-of-three generation strategy, and one Indigenous-oriented benchmark.

6.4 Transformer Adaptation for Community-Centered Moderation

The results also demonstrate the practicality of parameter-efficient Transformer adaptation for community-centered moderation. By combining LoRA adaptation with quantized local inference, Botochtone can operate without external cloud-based inference or full-model retraining. In the experimental setting considered here, the approach shows that a relatively small, carefully curated Indigenous-oriented dataset can contribute to the adaptation of a large Transformer model for a specialized moderation task.

This deployment model is particularly relevant for Indigenous organizations, where privacy, data sovereignty, and community governance influence technology adoption. Processing messages entirely on local infrastructure keeps moderation decisions under community control while reducing dependence on external service providers.

6.5 Research Implications

Rather than proposing a universal moderation solution, Botochtone demonstrates the feasibility of combining Transformer adaptation, culturally informed prompting, and community-specific dataset construction to address a documented gap in current moderation research. The methodology may provide a useful foundation for future investigations into culturally adapted moderation systems for other underrepresented communities.

7 Threats to Validity

Internal Validity. The training corpus integrates three complementary datasets with consistent binary labeling applied across all sources prior to training. The Indigenous-oriented examples were produced through an AI-assisted generation process subject to systematic manual review and validation; as with any curated corpus, real-world interactions may exhibit additional linguistic variation. AI-assisted generation was used because, to the best of our knowledge, no publicly available moderation corpus specifically captures Indigenous-directed online harm. The generation process was grounded in the categories of harm identified in the Canadian Heritage report, providing a culturally relevant basis for constructing the examples. However, manually validated synthetic examples cannot fully reflect naturally occurring community discourse and may omit linguistic phenomena such as code-switching, evolving slang, and other forms of variation found in real-world communication. Future work should therefore complement this initial dataset with ethically collected and community-informed real-world data.

The ablation study follows an incremental design in which each configuration introduces a specific adaptation component. While the individual effects of prompting and dataset adaptation can be observed through successive configurations, interactions between these components may also contribute to the final performance gains.

Construct Validity. Performance is assessed using four standard metrics Precision, Recall, Accuracy, and F1-score providing a comprehensive characterization of binary moderation behavior. The binary classification formulation reflects a deliberate design choice aligned with the real-time requirements of community moderation. Extending the framework to multi-class harm categorization represents a natural direction for future work.

External Validity. As discussed in Section 4, the prototype was integrated with Discord; broader cross-platform evaluation remains future work. Discord serves as the integration surface of the prototype, while the moderation model operates directly on text and returns a binary moderation decision. The experiments reported in this work evaluate this text classifier on the Indigenous-oriented benchmark rather than end-to-end moderation performance across multiple

platforms. Consequently, the results should not be interpreted as evidence of cross-platform performance.

The Indigenous-oriented dataset and evaluation benchmark are also limited in scale, with 100 training examples and 132 evaluation messages. This study is intended as an initial evaluation of culturally adapted moderation in the Canadian Indigenous context rather than as evidence of representativeness across the many distinct First Nations, Inuit, and Métis communities. Larger datasets developed and validated with Indigenous communities would be needed to assess how well the findings generalize across communities and linguistic contexts.

The Indigenous-oriented dataset is grounded in a Canadian Heritage policy report [3], reflecting documented forms of online harm in a Canadian context. Evaluation across a broader range of Indigenous communities and geographic regions would further extend the generalizability of the findings.

Human Oversight. Human review was used during the construction of the Indigenous-oriented dataset and during the semantic validation of adversarial rewrites. This differs from a human-in-the-loop deployment setting in which moderators review and potentially correct the system’s decisions during operation. Such an evaluation was outside the scope of the present study. Future work could evaluate Botochtone with Indigenous community moderators to examine how model decisions align with community judgements and how human review could be incorporated into the moderation process.

Adversarial Robustness. The adversarial evaluation considers a specific and limited threat model. It targets the 58 harmful messages correctly detected at baseline using a single attacker model in a grey-box, best-of-three setting without adaptive detector feedback. In this grey-box setting, the attacker receives only a natural-language description of the moderation policy and has no access to the detector weights, its exact system prompt, or detector feedback. Consequently, the reported 8.6% validated attack success rate characterizes robustness only under this restricted attacker knowledge and interaction model. It should not be interpreted as evidence of robustness against other attacker models, white-box attacks, adaptive or iterative attacks, or other evaluation benchmarks. Future work should extend the evaluation to stronger and more diverse threat models, including adaptive attacks, alternative attacker models, and, where appropriate, white-box settings.

Conclusion Validity. The evaluation covers four structured configurations assessed on a dedicated 132-message benchmark. Although performance does not increase monotonically across all configurations, the final system substantially outperforms all intermediate versions, providing evidence that the combination of Indigenous-oriented training data and the Indigenous-aware prompt improves performance under the evaluated setting.

Replication using larger Indigenous-oriented benchmarks and independent evaluation datasets would further strengthen the statistical basis of the reported findings.

8 Conclusion

This paper presented Botochtone, a Transformer-based moderation framework designed to improve online safety in Indigenous communities. Despite growing investment in LLM-based content safety systems [13,9,32], moderation frameworks specifically designed for Indigenous populations remain underexplored in the AI safety literature. By combining parameter-efficient adaptation of a Ministral-8B Transformer, Indigenous-aware prompting, and a purpose-built Indigenous-oriented dataset, the framework directly addresses this gap by targeting the detection of culturally specific forms of harmful content that are often absent from conventional moderation benchmarks.

The empirical evaluation showed that the largest performance improvement among the evaluated configurations occurs when Indigenous-oriented training examples are incorporated. The final configuration achieved an F1-score of 98.3% and an Accuracy of 98.5% on an Indigenous-oriented evaluation benchmark, substantially outperforming the unadapted baseline (F1: 83.9%) and configurations trained exclusively on generic harmful-content datasets. These results provide evidence that culturally adapted training data can improve moderation performance in the setting evaluated in this study. A complementary grey-box adversarial evaluation identified semantically valid evasions for 5 of the 58 baseline-detected harmful messages, corresponding to a validated attack success rate of 8.6%. These findings are specific to the Canadian Indigenous context and to the experimental settings considered in this work.

Beyond performance improvements, Botochtone demonstrates the feasibility of combining local Transformer inference, LoRA adaptation, and culturally informed dataset construction to support privacy-preserving and community-centered moderation.

Future work will focus on expanding the Indigenous-oriented dataset with community-validated examples, evaluating the framework across additional platforms and community contexts, and extending the approach toward multilingual and multi-class moderation settings. Further evaluation should also examine human-in-the-loop moderation with Indigenous community reviewers and stronger adversarial settings involving multiple or adaptive attackers. The same general methodology could also be investigated in other community-specific moderation contexts, provided that the relevant cultural characteristics and data-governance requirements are considered.

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Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Ainslie, J., Lee-Thorp, J., de Jong, M., Zemlyanskiy, Y., Lebrón, F., Sanghai, S.: GQA: Training generalized multi-query transformer models from multi-head checkpoints. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 4895–4901 (2023)
2. Arslan, M., et al.: Detecting LGBTQ+ instances of cyberbullying. arXiv preprint arXiv:2409.12263 (2024)
3. Bain, S., Paradis, G., Stockall, C.: What we heard report: Indigenous online safety (2023), <https://www.canada.ca>
4. Blodgett, S.L., Barocas, S., Daumé III, H., Wallach, H.: Language (technology) is power: A critical survey of “bias” in NLP. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics. pp. 5454–5476 (2020)
5. Cohen, J.: A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **20**(1), 37–46 (1960). <https://doi.org/10.1177/001316446002000104>
6. Gajewska, E., Budzyńska, K., Chudziak, J.A.: Improving implicit hate speech detection via a community-driven multi-agent framework. arXiv preprint arXiv:2601.09342 (2026)
7. Gerganov, G., llama.cpp contributors: llama.cpp: LLM inference in C/C++. <https://github.com/ggerganov/llama.cpp> (2023)
8. Girón, A., Huertas-Tato, J., Camacho, D.: LLM synthetic generation to enhance online content moderation generalization in hate speech scenarios. *Computing* **107** (2025). <https://doi.org/10.1007/s00607-025-01518-8>
9. Han, S., Kim, K., Yun, R., Singh, V., Srinivasan, M., Zhou, J., Choi, Y., Jiang, L.: WildGuard: Open one-stop moderation tools for safety risks, jailbreaks, and refusals of LLMs. arXiv preprint arXiv:2406.18495 (2024)
10. Hango, D.: Online harms among youth and young adults: The prevalence and nature of cybervictimization (2023), <https://www150.statcan.gc.ca>
11. Hartmann, D., Oueslati, A., Stauffer, D., Pohlmann, L., Munzert, S., Heuer, H.: Lost in moderation: How commercial content moderation APIs over- and under-moderate group-targeted hate speech and linguistic variations. In: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (2025). <https://doi.org/10.1145/3706598.3713998>
12. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: International Conference on Learning Representations (ICLR) (2022)

13. Inan, H., Upasani, K., Chi, J., Regan, R., Cho, M., Rezaii, T., Mehdad, Y., Meshkatiyan, E., Yin, X., Ozlem Bingol, S., et al.: Llama Guard: LLM-based input-output safeguard for human-AI conversations. arXiv preprint arXiv:2312.06674 (2023)
14. Jiang, A.Q., Sablayrolles, A., Mensch, A., et al.: Mistral 7b. arXiv preprint arXiv:2310.06825 (2023)
15. Kowalski, R.M., Giumetti, G.W., Schroeder, A.N., Lattanner, M.R.: Bullying in the digital age: A critical review and meta-analysis of cyberbullying research among youth. *Psychological Bulletin* **140**(4), 1073–1137 (2014)
16. Lees, A., Tran, V.Q., Tay, Y., Sorensen, J., Gupta, J., Metzler, D., Vasserman, L.: A new generation of perspective API: Efficient multilingual character-level transformers. In: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD). pp. 3197–3207 (2022). <https://doi.org/10.1145/3534678.3539147>
17. Mathew, B., Saha, P., Yimam, S.M., Biemann, C., Goyal, P., Mukherjee, A.: HateXplain: A benchmark dataset for explainable hate speech detection. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 35, pp. 14867–14875 (2021)
18. Mendu, S.K., Yenala, H., Gulati, A., Kumar, S., Agrawal, P.: Towards safer pretraining: Analyzing and filtering harmful content in webscale datasets for responsible LLMs. arXiv preprint arXiv:2505.02009 (2025)
19. Mistral AI: Ministraux: Ministral 3b and ministral 8b (2024), <https://mistral.ai/news/ministraux/>
20. Mozafari, M., Farahbakhsh, R., Crespi, N.: Hate speech detection and racial bias mitigation in social media based on BERT model. *PLoS ONE* **15**(8), e0237861 (2020). <https://doi.org/10.1371/journal.pone.0237861>
21. Muhammad, S.H., Abdulmumin, I., Ayele, A.A., Adelani, D.I., Ahmad, I., Aliyu, S.M., et al.: AfriHate: A multilingual collection of hate speech and abusive language datasets for african languages. arXiv preprint arXiv:2501.08284 (2025)
22. Park, J., Jeong, S., Song, S., Oh, A.: LLM-C3MOD: A human-LLM collaborative system for cross-cultural hate speech moderation. In: Proceedings of the Third Workshop on Cross-Cultural Considerations in NLP (C3NLP) (2025). <https://doi.org/10.18653/v1/2025.c3nlp-1.7>
23. Paul, J., Hong, F., et al.: Relations between bullying and distress among youth living in first nations communities: Assessing direct and moderating effects of culture-related variables. *Transcultural Psychiatry* **61**(3), 429–439 (2022)
24. Philipo, A.G., et al.: Cyberbullying detection: Exploring datasets, technologies, and approaches on social media platforms. *ACM Computing Surveys* (2025)
25. Sayed, F.R., Elnashar, E.H., Omara, F.A.: Cyberbullying detection in social media using natural language processing. *Scientific African* **28**, e02713 (2025)
26. Sokolova, M., Lapalme, G.: A systematic analysis of performance measures for classification tasks. *Information Processing & Management* **45**(4), 427–437 (2009). <https://doi.org/10.1016/j.ipm.2009.03.002>
27. Touvron, H., Lavril, T., Izacard, G., et al.: LLaMA: Open and efficient foundation language models. arXiv preprint arXiv:2302.13971 (2023)
28. Vanetik, N., Mimoun, E.: Detection of racist language in French tweets. *Information* **13**(7), 318 (2022). <https://doi.org/10.3390/info13070318>
29. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: *Advances in Neural Information Processing Systems*. vol. 30 (2017)

30. Vidgen, B., Thrush, T., Waseem, Z., et al.: Learning from the worst: Dynamically generated datasets to improve online hate detection. In: Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021. pp. 1667–1682 (2021)
31. Ye, H., Wisiosek, A., Maronikolakis, A., Alaccam, O., Schütze, H.: A federated approach to few-shot hate speech detection for marginalized communities. arXiv preprint arXiv:2412.04942 (2024)
32. Zeng, W., Liu, Y., Gelle, R., Chen, C., Shu, D., Singh, A., et al.: ShieldGemma: Generative AI content moderation based on gemma. arXiv preprint arXiv:2407.21772 (2024)